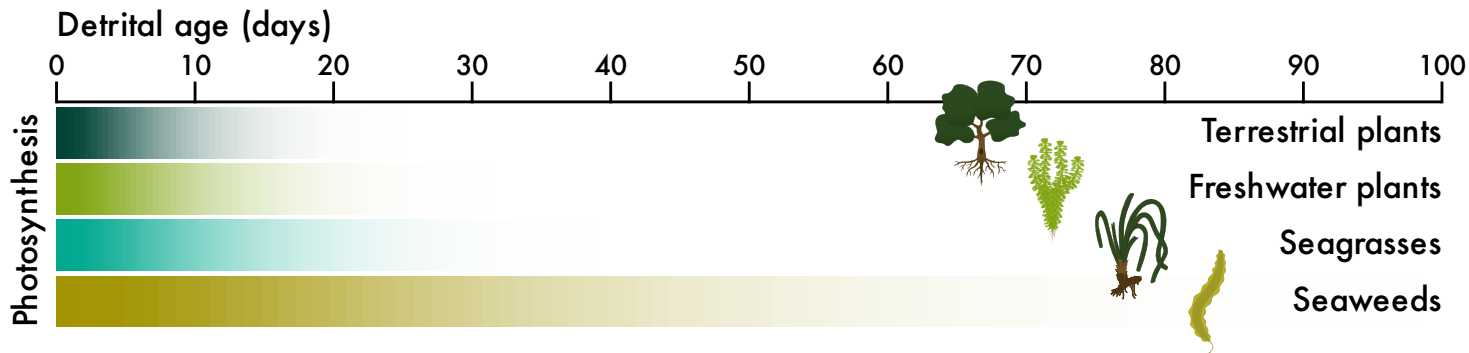
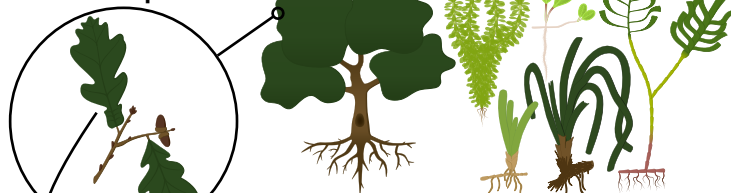
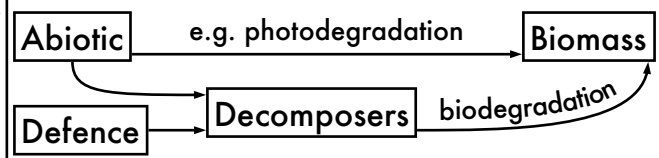


Vascular plants



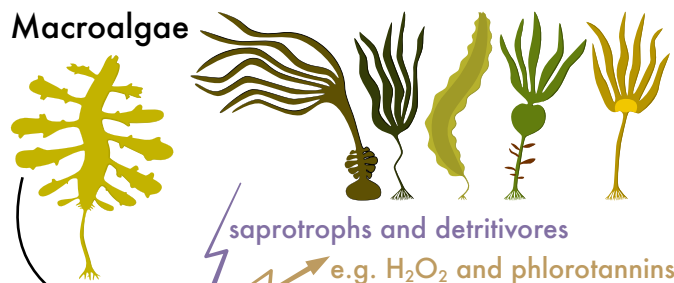
Conventional causal model



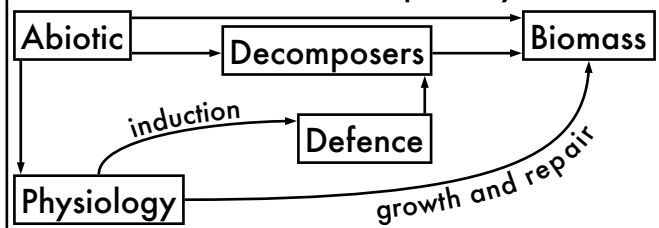
Physiologically independent recalcitrance dominates



Macroalgae



Causal model with detrital photosynthesis



saprotrophs and detritivores
e.g. H_2O_2 and phlorotannins

dead tissue

wound healing

Physiologically dependent recalcitrance dominates

Simultaneous decay of mass and physiology